

Informal learning in everyday human–LLM interaction

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Abstract

Large language models are increasingly used during everyday work, yet it remains unclear whether these interactions function only as tools for task completion or also as sites of informal learning. We analyse 142,825 naturalistic human–LLM conversations from Microsoft Copilot and WildChat, focusing on coding and writing as two common task ecologies. Across 563,614 user turns, cognitive engagement appeared in 30.3% of turns and constructive engagement in 4.9%, while emotional engagement was rare. Learning-oriented engagement was higher in intentional than in unintentional interactions and higher in coding than in writing, but it remained detectable even when conversations were not explicitly framed as learning. Conversations containing scaffolding support were consistently associated with richer constructive participation and deeper post-answer interaction, with stronger associations in coding than in writing. However, support was heterogeneous: coding conversations were dominated by explanation-heavy support, whereas writing conversations were dominated by more directive support, and support forms differed in their associations with constructive uptake. Temporal analyses showed that the clearest signal was local and state-dependent. Scaffolded assistant turns were more likely than direct-response turns to be followed by constructive user turns in most settings, but longer-span within-conversation contrasts were mixed. These findings identify everyday human–LLM interaction as an underexamined site of informal learning, while showing that learning-oriented engagement depends on task ecology, support form and temporal scale rather than on scaffolding presence alone.

Keywords: human–LLM interaction, informal learning, large language models, scaffolding, learner engagement

1 Introduction

Large language models (LLMs) are becoming embedded in the cognitive work of everyday life. People use them to write, code, search, troubleshoot and make sense of unfamiliar problems. Most of these interactions do not take place in classrooms, training programmes or purpose-built tutoring systems. Instead, they arise in the course of routine activity, when users turn to LLMs to solve an immediate problem, clarify a concept, improve a draft or navigate an unfamiliar task. In such moments, the model may do more than help complete the task at hand: it may shape what the user notices, understands and tries next. Everyday human–LLM interaction therefore constitutes a rapidly expanding setting for informal learning—that is, learning that occurs outside formal curricula and designed instructional environments, and instead emerges through everyday activity, self-directed inquiry and problem solving [1]. This learning may be intentional, when users explicitly seek understanding, or incidental, when understanding develops while pursuing another immediate goal [2, 3].

Despite the scale and growing ubiquity of such interactions, little is known about the human side of this emerging learning ecology: how often users move beyond obtaining an answer to engage with ideas in ways that support understanding, and which conditions make such engagement more likely. This question is fundamental because general-purpose AI systems increasingly give users immediate access to explanations, examples, critiques and feedback that may exceed their current expertise, at the very moment they are trying to act. As AI assistance becomes woven into work and study, learning from AI is unlikely to remain a special activity confined to tutoring applications; it will become part of how people routinely encounter unfamiliar knowledge. The critical issue is therefore not only whether LLMs can be engineered to provide instructional support, but whether everyday use elicits active and constructive engagement from users. Existing research on LLMs and learning has focused primarily on settings in which learning support is deliberately designed, such as automated feedback, question generation, retrieval practice and tutoring systems [4–7]. These studies demonstrate that LLMs can enact instructional strategies when they are explicitly prompted, trained or embedded in purpose-built systems [8]. But they leave a more basic question unresolved: when general-purpose LLMs are used in the course of everyday tasks, how often do users show learning-oriented engagement, and what factors are associated with deeper engagement?

Studying this question in naturalistic logs requires adapting educational theory to a setting for which it was not originally designed. Unlike classroom or tutoring data, everyday human–LLM conversations have no fixed instructional task, explicit learning objective or post-test outcome. A user may ask for a usable answer, request an explanation and test a revised understanding within the same exchange; an assistant response may likewise combine task completion with feedback, modelling, explanation, hints or questions. The learning-relevant signal is therefore not contained in any single

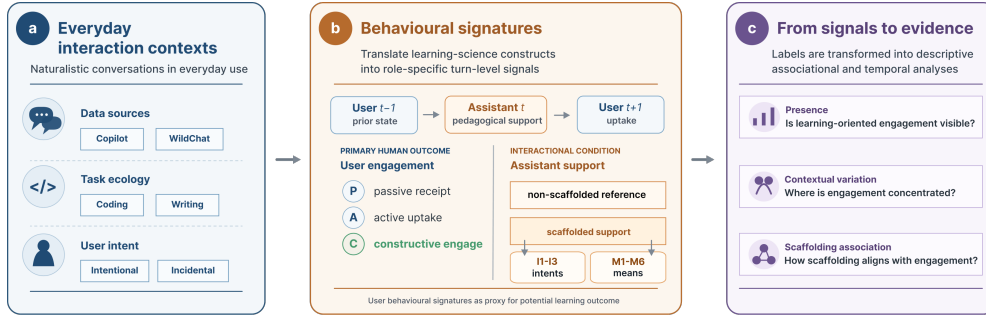


Fig. 1 Analytic framework for studying informal learning in everyday human-LLM interaction. **a).** Naturalistic conversations are analysed across data sources, task ecologies and user framing. **b).** Learning-science constructs are translated into role-specific turn-level behavioural signatures, with user engagement treated as the primary human outcome and assistant scaffolding as an interactional condition. **c).** The resulting labels support analyses of engagement presence, contextual variation and scaffolding-engagement association. These labels capture observable interactional signatures, not durable learning outcomes or causal instructional effects.

utterance alone, but in how user uptake and assistant support unfold across adjacent turns and longer conversational sequences. We therefore translate theories of engagement and scaffolding into role-specific, turn-level behavioural signatures: user turns are coded for passive, active or constructive uptake [9, 10], while assistant turns are coded for the presence of scaffolded support and further characterized by three support intents and six support means [11–14]. Following the literature that deeper forms of overt engagement are more strongly linked to learning [10], we treat constructive uptake as the strictest behavioural indicator of learning-oriented participation available in naturalistic logs, while recognizing that it is not a direct measure of knowledge gain. LLM-assisted annotation and human verification then allow this adapted framework to be applied at corpus scale. These measures do not establish causal learning effects or durable knowledge gains; instead, they provide a way to observe whether learning-oriented engagement is present in everyday use and to test which contextual and interactional conditions—including task ecology, user intent and assistant scaffolding—are associated with deeper user participation.

Using this framework, we analyse 142,825 naturalistic human-LLM conversations from Microsoft Copilot and WildChat, focusing on coding and writing as two common domains of everyday LLM use. The corpus contains 563,614 user turns and 561,698 assistant turns, allowing us to compare the prevalence and composition of engagement, variation across domain and learning-intent contexts, conversation-level associations between assistant support and user participation, differences among support forms, and local temporal coupling between adjacent turns.

We find that learning-oriented engagement is not exceptional but recurrent in everyday human-LLM interaction. Across the corpus, cognitive engagement appears in 30.3% of user turns, among which 16.2% is constructive level engagement, and assistant scaffolding appear in nearly 40% of assistant turns. These patterns show that everyday AI use often involves more than task completion alone: it creates repeated occasions for users to work with ideas, revise understanding and pursue sense-making.

Engagement is higher in intentional than in unintentional conversations and higher in coding than in writing, but it remains visible across all four matched settings, indicating that informal learning can emerge both as an explicit goal and as an incidental feature of practical work. Assistant scaffolding is associated with richer constructive participation, both across conversations and from one turn to the next, but this association is heterogeneous rather than universal. Support form, task ecology and temporal scale all matter: feedback-like and explanatory support show the clearest positive association with constructive engagement, whereas other forms vary across domains and platforms, and the strongest temporal evidence is local rather than cumulative. Together, these findings locate everyday human–LLM interaction in a middle space between ordinary tool use and designed instruction. General-purpose AI systems are not only changing how people complete routine tasks; they are also becoming part of the everyday interactional environments in which people encounter, test and extend understanding outside formal education.

2 Results

The analyses proceed from the presence of learning-oriented engagement to the conditions under which its constructive form appears. We first describe the prevalence and composition of user engagement, then focus on constructive engagement as the strictest observable indicator of learning-oriented participation, capturing turns in which users elaborate, test, revise or extend ideas. We ask how constructive engagement varies with user framing, task ecology and interaction depth, and whether assistant scaffolding is associated with richer constructive participation across conversations, support forms and adjacent turns. Because the logs contain no direct assessment of knowledge gain, engagement is interpreted as a behavioural signature of potential learning, and scaffolding analyses are interpreted as associations rather than causal evidence of learning effects.

2.1 Everyday human–LLM conversations contain learning-oriented engagement

Across the four dataset-by-task settings, 30.3% of user turns showed cognitive engagement, 4.9% reached constructive engagement, and emotional engagement was rare, appearing in only 0.62% of user turns (Table 1). This establishes the first descriptive pattern: everyday human–LLM interaction contains observable learning-oriented uptake, but that uptake is primarily cognitive rather than affective, and only a smaller subset reaches the more generative constructive form. In other words, users often did more than simply request or receive task output, but the depth of that engagement varied. We therefore first examine the internal structure of cognitively engaged turns before turning to the contexts in which constructive engagement is concentrated.

Figure 2a decomposes cognitively engaged turns into passive receipt, active uptake and constructive engagement. Active uptake accounted for 70.8% to 86.3% of cognitively engaged turns across the four dataset-by-task settings, constructive engagement for 12.6% to 23.5%, and passive receipt for 1.1% to 6.9%. Thus, learning-oriented engagement was not a binary signal but a depth-graded pattern: most cognitively

Table 1 Dataset and behavioural overview. Summary statistics for the four dataset-by-task settings analysed in the study. Constructive engagement is reported as the stricter depth level within cognitive engagement and serves as the focal user-side outcome in subsequent analyses.

Setting	Sample size			User framing	Assistant support	Cognitive engagement		Emotional engagement
	Conversations	User turns	Assistant turns	Intentional conv. (%)	Scaffolded turns (%)	Overall (%)	Constructive level (%)	Emotional (%)
WildChat coding	31,878	124,073	124,623	37.13	51.03	50.21	9.23	0.95
WildChat writing	39,534	163,705	164,824	25.80	23.53	16.34	2.22	0.57
Copilot coding	31,879	127,435	125,610	44.59	49.50	48.11	6.05	0.76
Copilot writing	39,534	148,401	146,641	36.98	41.00	13.88	3.26	0.30
Total / pooled	142,825	563,614	561,698	36.13	40.00	30.3	4.9	0.62

Note. Percentages for cognitive, constructive and emotional engagement are computed over user turns; percentages for scaffolded support are computed over assistant turns. The constructive-level column reports the share of all user turns that reached the constructive depth level within cognitive engagement. Figure 2a reports the composition of passive, active and constructive uptake among cognitively engaged turns.

engaged turns involved taking up or using the assistant’s response, whereas a smaller but stable subset involved users elaborating, testing, revising or extending ideas. Because this constructive subset is the strictest observable behavioural signal of learning-oriented participation in our framework, we use constructive engagement as the focal user-side outcome in the analyses that follow.

2.2 Constructive engagement is shaped by user framing, task ecology and interaction depth

Having established that everyday human–LLM conversations contain learning-oriented engagement, we next examined where its constructive form was concentrated. User framing provided the first contrast (Fig. 2b). Across all four dataset-by-task settings, conversations explicitly oriented toward learning or exploration showed higher cognitive engagement than conversations without such framing, with rates ranging from 26.0% to 76.6% in intentional conversations and from 6.4% to 33.2% in unintentional conversations. Constructive engagement showed the same directional pattern in every setting, increasing from 1.5–5.7% in conversations without explicit learning framing to 5.2–13.1% in intentional conversations. Explicit learning or exploratory framing therefore amplified constructive engagement, but did not create it from zero: even task-oriented conversations without explicit learning intent contained constructive participation.

Task ecology provided a second source of variation. In the two everyday task ecologies, coding-oriented conversations showed higher cognitive engagement than writing-oriented conversations in both datasets: 50.21% versus 16.34% in WildChat and 48.11% versus 13.88% in Copilot. Constructive engagement also differed by task ecology, with 6.1–9.2% in coding and 2.2–3.3% in writing. Importantly, this contrast was not reducible to explicit learning framing: in both datasets, unintentional coding conversations showed higher cognitive engagement than intentional writing conversations. This pattern suggests that task ecology was not merely a backdrop for explicit learning intent; it shaped the baseline opportunities for users to ask questions, test assumptions, refine solutions and extend ideas.

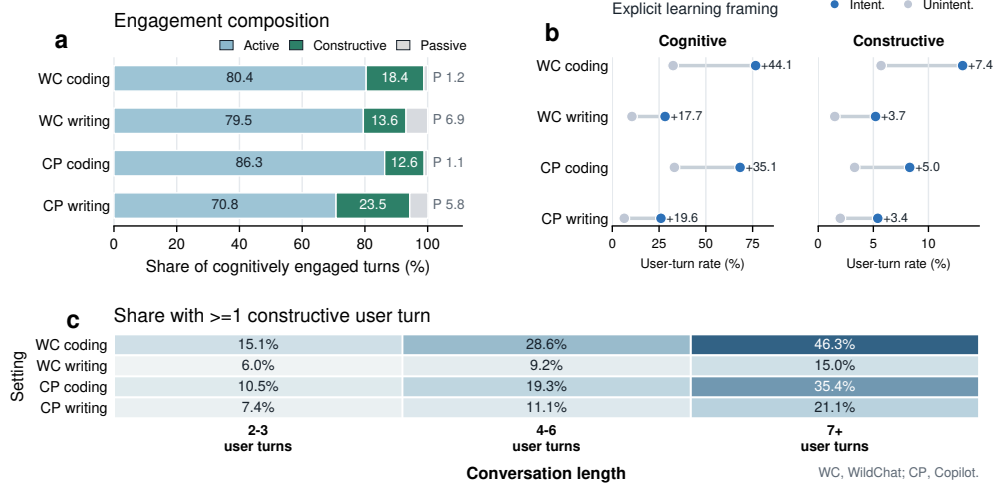


Fig. 2 Learning-oriented engagement varies by engagement form, user intent and interaction depth. **a**, Distribution of active, constructive and passive engagement within cognitively engaged user turns across the four matched settings. **b**, User-turn rates of cognitive and constructive engagement in intentional and unintentional conversations. **c**, Probability that a conversation contains at least one constructive user turn across behavioural-depth buckets. Panel a reports percentages within cognitively engaged turns; panel b reports percentages of user turns; panel c reports conversation-level probabilities.

Interaction depth provided a third lens on where constructive participation surfaced (Fig. 2c). Across all four dataset-by-task settings, the probability that a conversation contained at least one constructive user turn increased monotonically with behavioural depth, from 0.060–0.151 in 2–3-turn conversations to 0.150–0.463 in conversations with seven or more turns. The gradient was steepest in coding-oriented conversations, but the same direction appeared in writing-oriented conversations as well. Because this measure captures whether a constructive turn occurred at least once in a conversation, longer exchanges naturally provide more opportunities for such turns to appear and may also reflect more complex tasks or more engaged users. We therefore do not interpret depth as evidence that turns accumulate into learning progress by themselves. The substantive point is instead that constructive engagement is most visible when everyday LLM use moves beyond one-off answer delivery and becomes a sustained exchange in which users return to, interrogate and reshape the assistant’s response.

Together, these results show that constructive engagement is ecological. It is amplified by explicit learning framing, varies across task ecologies and concentrates in sustained exchanges. This motivates the next step of the analysis: asking whether assistant scaffolding is one interactional condition associated with richer constructive participation.

2.3 Scaffolded conversations show richer constructive participation

We next asked whether assistant scaffolding marked conversations with richer constructive participation. At the conversation level, interactions containing at least one scaffolded assistant turn showed higher constructive ratios than interactions without scaffolded support in all four dataset-by-task settings (Fig. 3a). The raw differences ranged from +1.1 to +3.9 percentage points, with larger contrasts in coding-oriented than in writing-oriented conversations. Scaffolded conversations were also deeper after the first assistant response, with post-answer depth differences ranging from +1.61 to +3.13 turns. These raw contrasts provide an initial signal that scaffolding co-occurred with richer constructive participation, but they are not self-explanatory. Because scaffolded support was not randomly assigned, conversations with scaffolding may also have been more complex, more explicitly learning-oriented or more likely to continue. Raw comparisons therefore mix the support–engagement association with differences in task ecology, user framing and interactional opportunity.

We therefore used covariate-adjusted models as a partial check on this compositional ambiguity[Cite:]. These models asked whether scaffolded-support presence remained associated with constructive participation after conditioning on observed contextual and interactional features. In Poisson models predicting constructive-turn counts per conversation, the expected constructive-turn multiplier associated with scaffolded support ranged from 1.240 to 1.870 across the four dataset-by-task settings. In logistic models predicting whether a conversation contained at least one constructive turn, the corresponding odds ratios ranged from 1.251 to 1.761 (Fig. 3b). The adjusted estimates were consistently positive, indicating that scaffolded support was not fully explained by measured differences in conversation composition. At the same time, the estimates differed across task ecologies, with stronger associations in coding-oriented conversations than in writing-oriented conversations, and should not be interpreted as causal effects.

The support–engagement association was also shaped by user framing. Stratified models showed stronger associations in intentional than in unintentional conversations in three of the four dataset-by-task settings; WildChat writing was the exception, where the intentional and unintentional estimates were nearly identical [Cite:]. This pattern echoes the context results above: scaffolding aligned most strongly with constructive participation when users were already approaching the exchange with explicit learning or exploratory intent. Scaffolding therefore did not operate as an isolated property of the assistant response. Its association with constructive engagement depended on the broader interactional state of the conversation.

We also used coding-oriented conversations to examine a learning-relevant form of sustained engagement: persistence in response to breakdown. In everyday informal learning, difficulty is a diagnostic moment because the user can either disengage, ask for another answer or continue working through the problem[Cite:]. Persistence after failure therefore complements constructive uptake: it captures whether the user remains in a repair-and-sense-making loop when the initial attempt does not work. This measure is most interpretable in coding, where breakdowns are more directly observable through errors, failed execution, debugging language and error-resolution

Table 2 Summary of key support–engagement effects across the four matched settings. Constructive-ratio contrasts compare conversations with versus without scaffolding support. Adjusted models report the association of having scaffolding support with constructive-turn count (Poisson rate ratio) and with having at least one constructive turn (logit odds ratio). Adjacent-turn lift is the difference in the probability that the next user turn is constructive after scaffolded versus direct-response assistant turns.

Setting	Constructive ratio Has S2 / No S2	Difference (pp)	Depth diff. (turns)	Poisson RR	Logit OR	Adjacent lift (pp)	Around first S2 diff.
WildChat coding	0.094 / 0.055	+3.9	+2.23	1.852	1.761	+2.68	+0.014
WildChat writing	0.032 / 0.020	+1.2	+3.13	1.569	1.437	+1.13	-0.042
Copilot coding	0.062 / 0.035	+2.7	+2.63	1.870	1.736	+1.65	-0.003
Copilot writing	0.037 / 0.026	+1.1	+1.61	1.240	1.251	+0.01	-0.039

contexts. In both coding datasets, scaffolded conversations showed substantially higher persistence after failure than conversations without scaffolded support: 0.895 versus 0.440 in WildChat coding and 0.759 versus 0.240 in Copilot coding (Fig. 3c). Thus, in a task ecology where breakdowns make the need for repair visible, scaffolded support was associated not only with richer constructive participation overall, but also with sustained user engagement when the task failed or became difficult.

Taken together, these results position scaffolded support as a meaningful interactional condition for constructive participation, but not as a stand-alone explanation for engagement. Scaffolded conversations were more constructive and deeper, the adjusted associations remained positive, and coding-oriented conversations showed greater persistence after observable breakdowns. Yet these patterns varied by task ecology and user framing. Scaffolding therefore appears most informative when considered as part of a broader interactional ecology: what users are trying to do, what the task makes visible and how support is delivered all shape whether users continue working constructively with ideas. This heterogeneity motivates a finer-grained analysis of scaffolding itself: whether different forms of support carry different signals for constructive engagement.

2.4 Feedback and explanation are most linked to constructive engagement

The conversation-level association did not imply that all forms of scaffolding carried the same engagement signal. We therefore decomposed scaffolded support into the six scaffolding-means labels in the annotation scheme (M1–M6) and compared scaffolded conversations containing each label with scaffolded conversations that did not contain that label, stratified by whether the conversation was intentionally or unintentionally learning-oriented. Because means labels were not mutually exclusive, these contrasts are descriptive associations rather than effects of isolated instructional moves. Even with this constraint, the pattern clearly differentiated support forms (Fig. 4a).

The overall cognitive-engagement panel shows that support forms differed in how they drew users into learning-oriented participation. Feedback and hinting were more strongly associated with cognitive engagement in unintentional conversations than in intentional conversations (M1: +16.7 versus +1.7 percentage points; M2: +14.2 versus +8.4 points). This pattern suggests that evaluative feedback and partial cues may

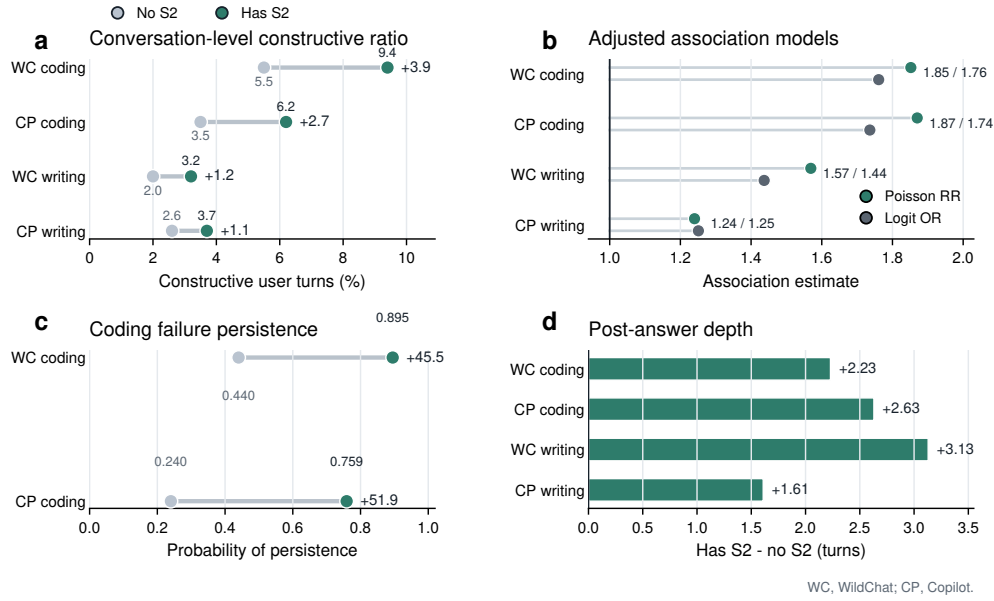


Fig. 3 Scaffolded conversations are associated with richer constructive participation and greater persistence in some settings. **a**, Conversation-level constructive ratios in conversations with versus without at least one scaffolded assistant turn. **b**, Adjusted model estimates for the association between scaffolded-support presence and constructive participation, reported as Poisson rate ratios for constructive-turn counts and logistic odds ratios for whether a conversation contained at least one constructive turn. **c**, Probability of persistence after failure in coding-oriented conversations with versus without scaffolded support. Persistence after failure is reported only for coding-oriented conversations because failure markers are more directly observable in debugging and error-resolution contexts. **d**, Difference in post-answer interaction depth between scaffolded and non-scaffolded conversations, reported as additional turns after the first assistant response. Numbers beside panels a, c and d indicate differences. WC, WildChat; CP, Copilot.

be especially useful for pulling otherwise task-oriented exchanges into cognitive work. Explanation and modelling, by contrast, showed larger cognitive-engagement contrasts in intentionally framed conversations (M4: +27.0 versus +16.7 points; M5: +11.4 versus +9.9 points), consistent with these forms aligning with users who are already seeking understanding. Instructing and questioning showed a different profile: instructing was negatively associated with cognitive engagement in both framing groups, and questioning was negative in intentional conversations and near zero in unintentional conversations. A plausible interactional explanation is that detailed instructions can make the next step executable without requiring users to articulate or revise their understanding, while questions in everyday assistant use may often function as clarification requests rather than designed Socratic prompts.

The constructive-engagement panel sharpened this distinction by separating broad cognitive uptake from more generative participation. Feedback-like support showed the strongest constructive-engagement contrast, especially in intentionally framed conversations (M1: +12.7 versus +5.7 percentage points). This is notable because feedback

did not show the same pattern for overall cognitive engagement in intentional conversations, suggesting that its contribution is not simply to increase participation in general. Rather, feedback may be especially aligned with constructive engagement because it responds to a user’s prior attempt and creates an occasion for revision, evaluation or extension. Explanation also showed positive constructive-engagement contrasts (M4: +6.0 versus +3.1 points), consistent with the idea that making mechanisms visible can support users in elaborating or testing their understanding. Hinting and modelling were positive but smaller, suggesting that partial cues and examples may support progress without always eliciting visible elaboration. By contrast, instructing and questioning were associated with lower constructive engagement in both framing groups, with stronger negative contrasts in intentionally framed conversations (M3: -4.9 versus -1.2 points; M6: -2.4 versus -0.7 points). Thus, the support forms most aligned with constructive engagement were those that preserved room for users to evaluate, revise and extend ideas, rather than those that primarily prescribed the next step or shifted the exchange into clarification.

Assistant support-intent distinctions were less discriminating. Most scaffolded support was cognitively oriented: I2-only support accounted for 91.9% to 95.4% of scaffolded turns across the four matched settings, whereas metacognitive and emotional support were much rarer. Intent categories therefore captured a high-level orientation of support, but they did not separate the interactional moves most informative for constructive engagement. The more useful distinction was how support was delivered.

Support supply made these form-level signatures consequential for task ecology (Fig. 4b). Coding-oriented scaffolded turns were dominated by explanation, with M4 appearing in 84.0% of scaffolded WildChat coding turns and 77.5% of scaffolded Copilot coding turns. Writing-oriented scaffolded turns were dominated by instructing, with M3 appearing in 59.2% of scaffolded WildChat writing turns and 66.5% of scaffolded Copilot writing turns. This distribution helps explain why task ecology matters for learning-oriented engagement. Users in different task settings did not merely receive more or less scaffolding; they encountered different mixtures of support. In the sampled coding ecologies, scaffolded support more often took the form of explanation, which was aligned with constructive sense-making. In the sampled writing ecologies, scaffolded support more often took the form of instruction, which may support task completion but was less aligned with visible constructive elaboration.

Together, these results show why scaffolding presence alone is too coarse to characterize learning-oriented human–LLM interaction. Scaffolded support was associated with richer constructive participation at the conversation level, but that association was carried unevenly by support form, user framing and task-specific support supply. The implication is not that systems should simply provide more scaffolding, but that learning-oriented support needs to be matched to the user’s current framing and the task ecology: feedback and explanation may sustain users who are already working with ideas, whereas more directive forms may complete the task without eliciting further elaboration.

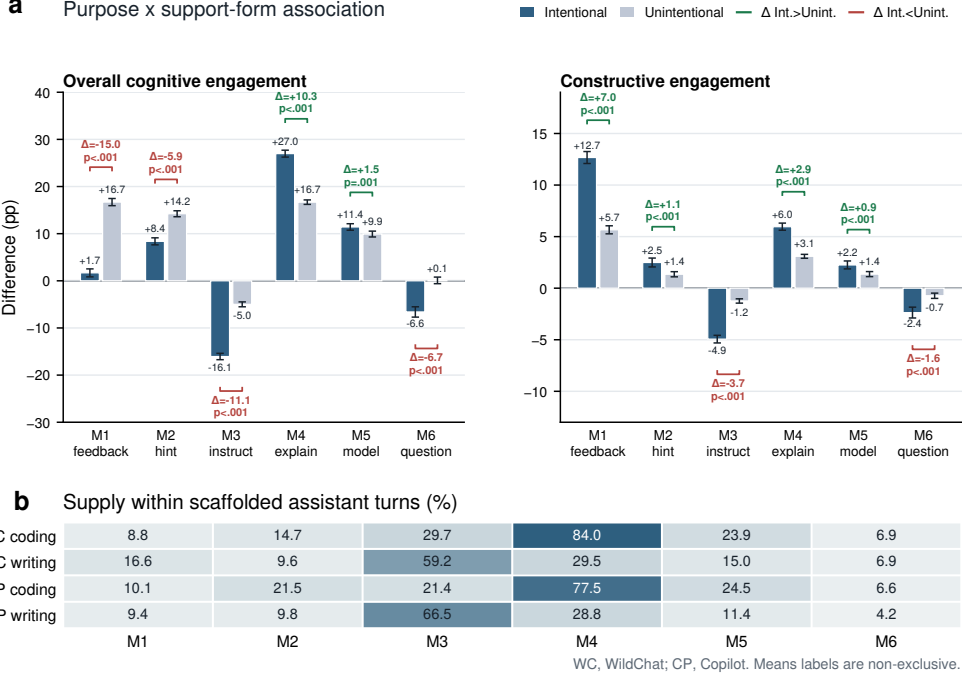


Fig. 4 Support form, user purpose and support supply differentiate scaffolded interaction. **a**, Conversation-level engagement differences associated with each scaffolding-means label within scaffolded conversations, pooled across the four matched settings and stratified by intentional versus unintentional learning orientation. Left and right panels report overall cognitive engagement and constructive engagement, respectively. Bars compare conversations containing each means label with scaffolded conversations that did not contain that label; intervals show 95% confidence intervals. Brackets report the intentional-unintentional difference in the means-label association and the interaction p value; green brackets indicate larger intentional contrasts and red brackets indicate larger unintentional contrasts. **b**, Percentage of scaffolded assistant turns containing each scaffolding-means label by setting. Means labels are not mutually exclusive, so rows do not sum to 100%. WC, WildChat; CP, Copilot.

2.5 Support and constructive engagement are locally, not cumulatively, coupled

Having shown that scaffolding varied by support form, user framing and task-specific supply, we finally asked whether the support-engagement relationship was visible in the temporal ordering of turns. The clearest signal appeared at the adjacent-turn level. When assistant turns contained scaffolded support rather than falling in the non-scaffolded reference category, the next user turn was more likely to be constructive in three of the four dataset-by-task settings (Fig. 5a). The lift was strongest in coding-oriented conversations (+2.68 percentage points in WildChat coding and +1.65 points in Copilot coding), positive but smaller in WildChat writing (+1.13 points), and effectively absent in Copilot writing. Thus, the temporal evidence was strongest where

scaffolding and constructive participation were both more prominent, and it appeared locally around adjacent turns rather than as a broad conversation-wide trend.

This local association depended on the user’s immediately preceding state (Fig. 5b). In both coding settings, scaffolded support was followed by higher constructive engagement across prior constructive, active and passive user turns. The largest coding lifts appeared after prior constructive turns in WildChat coding (+4.01 percentage points) and after prior passive turns in Copilot coding (+3.66 points). Writing-oriented conversations showed a less stable pattern: WildChat writing showed its clearest lift after an already constructive user turn (+6.27 points), whereas Copilot writing was neutral overall and positive only after passive prior turns. These patterns suggest that scaffolding did not have a single local value. Depending on the prior state, it could help sustain an already constructive thread or create an opportunity for renewed engagement after a less elaborative turn.

The reverse ordering showed the same state dependence (Fig. 5c). In coding-oriented conversations, assistant scaffolding was more likely to follow constructive or active user turns than passive turns: the probability of scaffolded support after constructive turns was 65.5% in WildChat coding and 64.3% in Copilot coding, compared with 29.5% and 23.1% after passive turns. Writing-oriented conversations also showed sensitivity to user state, but without the same ordering across datasets. This reverse contingency is important because it shows that assistant support was not simply an external input applied to otherwise stable conversations. User engagement state and assistant scaffolding were locally coupled: users’ current participation shaped the support they received, and scaffolded support in turn aligned with subsequent constructive engagement.

Coarser temporal contrasts were less consistent (Fig. 5d). Comparing constructive engagement after versus before the first scaffolded assistant turn yielded a small positive contrast in WildChat coding (+1.4 percentage points), but near-zero or negative contrasts in Copilot coding (−0.3 points), WildChat writing (−4.2 points) and Copilot writing (−3.9 points). These before–after contrasts are not equivalent to treatment effects: the first scaffolded turn may occur after difficulty, uncertainty or an already developing exchange. A different longer-span lens in WildChat showed increases in constructive engagement from the first to the second half of conversations in both coding (0.0725 to 0.0891, $p < 0.001$) and writing (0.0215 to 0.0288, $p < 0.001$). The mixed pattern therefore argues against a simple cumulative-uplift account in which scaffolded support steadily raises constructive engagement over the remainder of a conversation.

Together, the temporal analyses sharpen the interpretation of scaffolding. Scaffolded support was associated with constructive participation, but the strongest evidence was local, state-dependent and task-specific. Everyday human–LLM interaction therefore looks less like a one-way instructional pipeline and more like a coupled interaction process, in which support and user engagement co-evolve across adjacent turns without necessarily accumulating into a uniform upward trajectory.

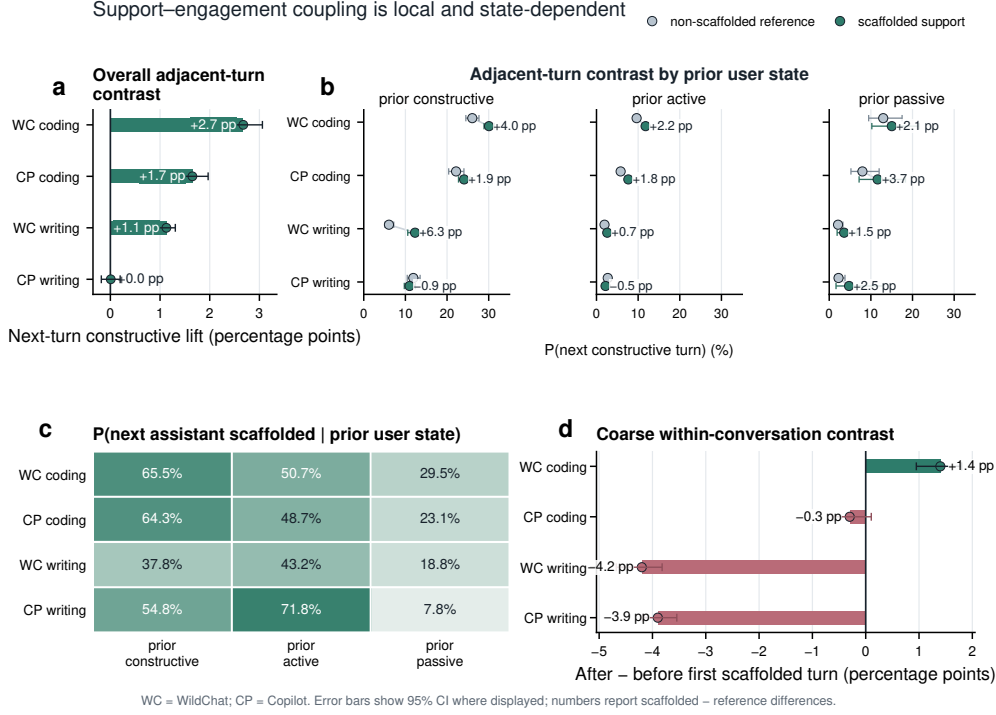


Fig. 5 Support–engagement coupling is local and state-dependent. **a**, Overall difference in the probability that the next user turn is constructive after scaffolded assistant turns versus non-scaffolded reference turns across dataset-by-task settings. **b**, Probability that the next user turn is constructive after scaffolded versus non-scaffolded reference assistant turns, stratified by the immediately preceding user state. Numbers indicate scaffolded-minus-reference differences in percentage points. **c**, Probability that the next assistant turn contains scaffolded support, conditional on the preceding user state. **d**, Coarse within-conversation contrast in constructive engagement after versus before the first scaffolded assistant turn. Panels **a** and **d** report percentage-point differences; panel **b** reports probabilities with percentage-point lift annotations; panel **c** reports probabilities. Error bars indicate 95% confidence intervals where shown. WC, WildChat; CP, Copilot.

2.6 WildChat model snapshots preserve the main pattern

As a robustness check, we repeated the main WildChat analyses across user-facing model snapshots and coarse model families. These analyses preserved the main directional patterns: coding generally showed higher cognitive engagement than writing, intentionality gaps were visible across most user-facing models, conversation-level support associations were broadly positive, and adjacent-turn signals were more interpretable than coarser within-conversation contrasts. We treat these analyses as robustness checks rather than evidence for model-specific mechanisms, because model labels in WildChat are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

3 Discussion

This study identifies an everyday interactional route through which human–LLM use can become a setting for informal learning. Learning-oriented engagement was common across ordinary coding and writing conversations, but its constructive form was selective: it concentrated in explicit learning intent, coding tasks and deeper exchanges. Assistant scaffolding marked conversations with richer constructive participation, yet the association depended on support form and temporal scale. Feedback-like and explanatory moves were more consistently linked to constructive engagement than directive or questioning moves, and the strongest temporal evidence appeared locally, from one turn to the next, rather than as a simple cumulative rise across a conversation. Together, these findings show that informal learning in everyday LLM use is not a generic consequence of exposure to AI; it emerges when users and assistants sustain iterative work with ideas.

The main finding is therefore not simply that engagement exists, but that constructive engagement is ecological. It is amplified by explicit learning intent, strongly structured by task domain and concentrated in deeper interactional sequences. This depth pattern should be read as an opportunity marker: longer conversations provide more occasions for constructive turns and may reflect more complex tasks or more engaged users, but they do not by themselves imply a cumulative learning trajectory. Coding produced substantially more cognitive and constructive engagement than writing, even when conversations were not explicitly framed as learning. This suggests that task ecology can shape the baseline opportunity for informal learning as much as users’ stated intentions do. For research on human–LLM interaction, the implication is that everyday LLM use should not be treated as a single behavioural environment. Different domains organize different opportunities for users to ask, test, refine and extend ideas.

Assistant scaffolding was associated with richer constructive participation, but the association was heterogeneous rather than treatment-like. Conversations containing scaffolded support had higher constructive ratios and greater post-answer depth across all four matched settings, and adjusted models showed that these associations were not explained away by observed covariates. In coding, scaffolded conversations also showed higher persistence after difficulty. These results suggest that scaffolding is part of the interactional environment in which constructive participation becomes more likely. At the same time, the observational design prevents causal claims about instructional efficacy. The contribution is to identify structured associations in naturalistic use, not to show that exposure to scaffolding produces durable learning gains.

A central implication is that increasing engagement is not a matter of increasing scaffolding in general. Support form mattered. Feedback-like and explanatory moves were the most consistently positive forms for constructive engagement, whereas instructing and questioning were weaker or negative. Support supply also differed by domain: coding conversations were explanation-heavy, while writing conversations were more directive. This two-sided heterogeneity helps explain why support did not have a single portable engagement value. The same broad category of scaffolding can invite different forms of user participation depending on how it is delivered and on the task ecology in which it appears.

The temporal analyses further suggest that informal learning with LLMs is best understood as a local interaction process. Scaffolded turns were followed by higher constructive uptake in most settings, and users’ current engagement state shaped the probability of receiving subsequent scaffolding. Support and engagement therefore appeared as coupled states, not as independent events in a one-way instructional pipeline. This temporal pattern also guards against over-reading the conversation-level associations: the data show local reciprocal ordering, not experimentally identified amplification. Longer-span trajectories were more mixed: around-first-scaffold contrasts were small or negative in several settings, whereas first-half versus second-half analyses in WildChat showed increases in constructive engagement. The strongest temporal claim is therefore scale-specific. Everyday LLM interaction repeatedly creates local opportunities for constructive uptake, but those opportunities do not automatically accumulate into a uniform upward trajectory.

These findings clarify why everyday LLMs are neither merely tools for task completion nor full substitutes for designed instruction. In ordinary use, the same system may act as an assistant, explainer, feedback provider or directive instructor within a single conversation. The learning relevance of the interaction depends on whether these roles recruit further user thinking rather than simply resolving the immediate task. This perspective also explains why constructive engagement is a useful organizing outcome. It captures the moments when users are not only receiving information but actively working with it: testing a boundary, revising an idea, articulating a constraint or extending an explanation.

For design, the results point toward adaptive support that is calibrated to user state, task ecology and support form. Systems that aim to support informal learning should not simply add more instructional language. They should identify when a user is already constructively engaged and sustain that trajectory, when a user is active but not yet constructive and offer feedback or explanation that invites elaboration, and when a user primarily needs direct help without forcing a pedagogical detour. The domain contrast is important here. Coding often benefited from explanatory support and showed stronger persistence effects, whereas writing contained more directive support and weaker temporal gains. Future systems may therefore need domain-sensitive support policies rather than a single generic tutor mode.

Several limits define the boundary of the claims. First, the analyses are observational, so scaffolding should be interpreted as associated with engagement rather than causally responsible for it. Second, the labels capture behavioural manifestations of learning-oriented engagement rather than verified learning outcomes or durable knowledge change. Third, fine-grained support categories are noisier than the top-level distinction between direct response and scaffolding, and means labels are not mutually exclusive. Fourth, WildChat model labels provide useful robustness checks but are not experimental assignments; model comparisons may be confounded with user population, time and task mix. These limits are important, but they do not erase the central pattern: everyday human–LLM interaction contains structured, context-dependent dynamics characteristic of informal learning.

The most direct unresolved threat is selection into support: more complex tasks or more motivated users may both elicit scaffolding and produce constructive follow-up.

The adjacent-turn analyses reduce this concern by conditioning on the immediately preceding interaction state, but they do not eliminate unobserved time-varying confounding. The safest interpretation is therefore local, state-dependent coupling between assistant support and user engagement rather than cumulative uplift or causal learning gain.

Overall, the study identifies a middle space in contemporary AI use. LLMs are not only instruments for faster task completion, and they are not only educational technologies when explicitly deployed as tutors. They are increasingly part of the ordinary interactional settings in which people clarify, test, revise and extend understanding while working. Studying this middle space is necessary for designing future systems that support learning without assuming that every user has entered a classroom, and for understanding how general-purpose AI may reshape informal learning at scale.

4 Methods

4.1 Study design

This study is an observational analysis of naturalistic human–LLM conversations. It compares Microsoft Copilot and WildChat across two everyday task ecologies, coding and writing, and asks whether learning-oriented engagement is visible in ordinary use, where constructive engagement is concentrated and how assistant scaffolding is ordered with user engagement across conversation-level and adjacent-turn timescales. The design is descriptive and associational. It does not estimate randomized exposure effects, causal instructional efficacy or durable learning outcomes.

The empirical sequence mirrors the Results section. We first estimate the prevalence and composition of user engagement, then examine how constructive engagement varies by user framing, task ecology and interaction depth. We next test whether scaffolded assistant support marks conversations with richer constructive participation, separate scaffolded support by support form and analyse local temporal coupling between assistant support and user engagement. Dataset and task ecology are treated as substantive dimensions of the interaction environment rather than nuisance variables to be averaged away.

4.2 Data sources and analytic samples

The analytic corpus contains 142,825 conversations and 1,125,312 total turns, including 563,614 user turns and 561,698 assistant turns. The four platform-by-task settings are WildChat coding, WildChat writing, Copilot coding and Copilot writing. WildChat contributes 31,878 coding conversations with 248,696 turns and 39,534 writing conversations with 328,529 turns. Copilot contributes 31,879 coding conversations with 253,045 turns and 39,534 writing conversations with 295,042 turns.

The matched settings were analysed with a common Level 1–4 annotation and analysis framework so that prevalence, contextual variation, support–engagement association and temporal ordering were computed from the same construct definitions. The manuscript reports the refreshed production outputs after the latest user-turn update

pass rather than earlier pre-update batches. Supplementary Section A.1 describes the corpus construction and refresh provenance in more detail.

4.3 Conversation metadata and task ecologies

Each conversation is represented as an ordered sequence of user and assistant turns together with available metadata. The main contextual variables are data source, task domain, learning intent, topic and language. Coding and writing were selected because they are common everyday LLM use cases but differ in their interactional affordances: coding often involves debugging, error diagnosis and iterative testing, whereas writing often involves drafting, revision and directive editing. Learning intent distinguishes conversations explicitly oriented toward understanding, exploration or learning from conversations in which learning-oriented engagement may be incidental to task completion.

Behavioural depth is measured from the number and ordering of turns in the conversation. Depth is used descriptively, as a marker of interactional opportunity and sustained exchange, not as an independent manipulation. Post-answer depth is measured as the number of turns after the first assistant response.

4.4 Turn-level constructs

The unit of annotation is the conversational turn. User and assistant turns are labelled with different constructs because they play different roles in the interaction. User turns are labelled for behavioural, emotional and cognitive engagement. The main analyses emphasize cognitive engagement and its depth levels. Passive engagement (P) captures reception or acknowledgement without substantial transformation. Active engagement (A) captures task-relevant action, follow-up or manipulation of supplied information. Constructive engagement (C) captures elaboration, revision, testing, extension or transformation of ideas. Constructive engagement is treated as the strictest behavioural proxy for learning-oriented participation available in these logs, not as a direct measure of knowledge gain.

Assistant turns are first distinguished as S1 or S2. S1 denotes a non-scaffolded reference response, including straightforward task fulfilment or stand-alone information provision. S2 denotes scaffolded support: assistant behaviour that structures, regulates or redirects the user’s process while leaving some cognitive work with the user. This definition follows the scaffolding tradition, especially contingency, transfer of responsibility and potential fading [11, 12, 14]. Scaffolded turns can also receive intention labels (I1 metacognitive, I2 cognitive and I3 affective support) and non-mutually exclusive means labels (M1 feedback, M2 hinting, M3 instructing, M4 explaining, M5 modelling and M6 questioning). Operational examples are provided in Supplementary Table C1.

4.5 Annotation workflow and verification

The annotation workflow combined theory-grounded codebook development, prompt iteration, parser checks, human review and production-scale LLM-assisted labelling. Prompt and post-processing rules were first developed on validation batches to reduce common failure modes, including over-labelling brief acknowledgements as active

engagement and misclassifying directive assistant responses as scaffolding. After the pipeline stabilized, the same annotation logic was applied to the large Copilot and WildChat corpora. All examples reproduced in the manuscript are de-identified and paraphrased rather than copied from raw logs.

Human verification was used as a quality-control input rather than as a separate source of outcome measurement. The final coding validation rerun combines three review rounds, with 658 task turns and 643 turns after deduplication. The final writing validation used 210 reviewed conversations. Agreement was summarized with per-label F1, Matthews correlation coefficient and Gwet’s AC1 because the labels are sparse, prevalence-imbalanced and not mutually exclusive in all cases. Agreement was strongest for constructive, passive, emotional and support-form labels; boundary-heavy labels such as writing A and S1 were less stable. Supplementary Table C2 reports the per-label agreement metrics, and Supplementary Section A.4 describes the final user-turn update check.

4.6 Outcome construction

The primary user-side outcome is constructive engagement. At the turn level, this is a binary indicator for whether the user turn is labelled C. At the conversation level, we use both the count of constructive user turns and an indicator for whether a conversation contains at least one constructive user turn. Constructive ratios are defined as constructive user turns divided by user turns.

The primary assistant-side exposure is scaffolded support. At the turn level, this is the distinction between S2 scaffolded support and S1 non-scaffolded reference response. At the conversation level, we use an indicator for whether a conversation contains at least one S2 assistant turn. Support-form analyses condition on scaffolded turns and compare conversations containing each means label with scaffolded conversations that do not contain that label. Because means labels can co-occur, these are descriptive support-form contrasts rather than mutually exclusive treatment comparisons.

4.7 Statistical analyses

Descriptive analyses estimate engagement prevalence, scaffolding prevalence, passive/active/constructive composition, emotional-engagement rates and behavioural-depth summaries within each platform-by-task setting. Context analyses compare intentional versus unintentional conversations and coding versus writing settings. Depth analyses group conversations into behavioural-depth buckets and estimate the probability that a conversation contains at least one constructive turn.

Conversation-level associations are analysed by comparing conversations with and without at least one scaffolded assistant turn. Constructive-turn counts are modelled with Poisson generalized linear models. Binary outcomes, including whether a conversation contains any constructive turn, whether the next user turn is constructive and whether the next assistant turn is scaffolded, are modelled with logistic regression or reported as conditional probability contrasts, depending on the estimand. Adjusted models include observed covariates appropriate to the analysis level, including learning intent, task domain or topic indicators, turn counts, emotional ratios, error-related

variables and turn index. Supplementary Tables C3–C5 summarize the main model families and temporal estimands.

Temporal analyses use two complementary scales. Adjacent-turn analyses compare the probability that the next user turn is constructive after scaffolded versus non-scaffolded assistant turns and the probability that the next assistant turn is scaffolded after different prior user states. Coarser within-conversation analyses compare constructive engagement before versus after the first scaffolded assistant turn and first-half versus second-half summaries where available. The adjacent-turn analyses test local ordering; the coarser analyses test whether there is evidence for longer-span accumulation.

4.8 Robustness, privacy and LLM use

WildChat robustness analyses repeat the main patterns across user-facing model snapshots and coarse model families. These checks assess whether the headline patterns are reducible to a single WildChat model label. They are not interpreted as model-causal comparisons, because model labels are not experimentally assigned and may be confounded with time, user population, task mix and logging context.

The analyses are bounded by the observational design and by the nature of the labels. They support claims that everyday human–LLM conversations contain measurable learning-oriented engagement; that constructive engagement varies by intent, task ecology and depth; that assistant scaffolding is associated with richer constructive participation; and that support and engagement show local reciprocal ordering. They do not support claims that scaffolded support causes durable learning gains, that all constructive engagement reflects actual knowledge change or that the labelled conversational states perfectly measure learning itself.

LLMs were used as objects of study and as annotation tools. Production labels were generated with LLM-assisted workflows and then checked through parser rules, targeted update scripts and human validation. The authors also used LLM-based tools to assist with code drafting, debugging, analysis workflow organization and language editing. LLM-generated assistance was not used as a substitute for human verification of the validation samples or for interpretation of the statistical results.

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Supplementary information

Supplementary information is provided below.

Appendix A Supporting methods

A.1 Corpus construction and matched analytic settings

The analyses use four platform-by-task settings: WildChat coding, WildChat writing, Copilot coding and Copilot writing. These settings were chosen to compare a public corpus with a proprietary product corpus while holding task ecology constant at a coarse level. Coding and writing were selected because both are common everyday LLM use cases, yet they differ in the opportunities they create for visible learning-oriented engagement. Coding conversations often contain error messages, debugging attempts, execution failures and iterative repair. Writing conversations more often involve drafting, rewriting, shortening, tone adjustment and direct editing.

The matched analytic corpus contains 142,825 conversations. WildChat contributes 31,878 coding conversations and 39,534 writing conversations; Copilot contributes 31,879 coding conversations and 39,534 writing conversations. Matching was used to keep the main comparisons focused on platform-by-task settings rather than on large differences in corpus size. All reported analyses use the refreshed Level 1–4 outputs after the latest user-turn update pass.

For each conversation, the pipeline retained the ordered turn sequence, role labels, available topic and language metadata, learning-intent indicators and model metadata where available. The model metadata are used only for robustness checks in WildChat because model assignment in naturalistic public conversations is observational and may reflect time, user population, task mix and logging context.

A.2 Annotation ontology

The codebook translates learning-science constructs into role-specific behavioural signatures. User turns are labelled for engagement. Passive engagement (P) marks acknowledgement, reception or minimal continuation without visible transformation. Active engagement (A) marks task-relevant action or follow-up, such as asking for a reformatted answer or applying supplied information. Constructive engagement (C) marks deeper work with ideas, such as testing an implication, revising a hypothesis, articulating a constraint, transferring a mechanism or extending an explanation. Emotional engagement is labelled separately because affective expressions were rare and not the main analytic outcome.

Assistant turns are labelled for support. S1 denotes a non-scaffolded reference response: the assistant provides the requested information or deliverable without materially structuring the user’s process. S2 denotes scaffolded support: the assistant structures, regulates, diagnoses or redirects the user’s work while leaving some responsibility with the user. Scaffolded turns can also receive support-intent labels and support-means labels. Intention labels distinguish metacognitive support (I1), cognitive support (I2) and affective support (I3). Means labels distinguish feedback (M1), hinting (M2), instructing (M3), explaining (M4), modelling (M5) and questioning

(M6). Means labels are non-mutually exclusive because a single assistant turn can both explain a mechanism and provide an example or hint. Supplementary Table C6 reports the support-means supply profile across the four platform-by-task settings.

A.3 LLM-assisted annotation pipeline

The annotation pipeline combined prompt-based labelling with deterministic parser checks and targeted post-processing. During development, prompts were iterated against validation examples and disagreement cases. Common failure modes included treating brief acknowledgements as active engagement, treating any long assistant answer as scaffolding, collapsing directive instructions into scaffolding without checking whether responsibility remained with the user, and over-labelling sparse support means.

Production labels were generated at turn level. Parser checks enforced the expected label schema, normalized label variants and flagged missing or malformed outputs. Post-processing rules were used only when a recurring error pattern could be specified transparently, such as a narrow class of user-turn false positives. Prompt versions, run metadata, disagreement files and update outputs were retained with the analysis artifacts so that validation cases can be traced without reproducing raw conversation text in the manuscript.

A.4 Human verification and update provenance

Human verification was used to quantify label reliability and identify boundary cases. Coding validation combined three review rounds, with 658 task turns and 643 turns after deduplication. Writing validation used 210 reviewed conversations. Because many labels are sparse or prevalence-imbalanced, agreement is reported with per-label F1, Matthews correlation coefficient and Gwet’s AC1 rather than with a single global agreement statistic.

The final user-turn check contained 100 WildChat coding examples and 100 WildChat writing examples reviewed after the update pass. These examples were used to identify residual false positives and calibrate patch-style update filters, especially for Active and Passive user-turn labels. The last WildChat user-turn update pass scanned 31,879 coding conversations and 39,534 writing conversations, applying 1,815 coding updates and 21 writing updates. The main manuscript reports the refreshed labels produced by this final pass.

A.5 Outcome construction

At the turn level, the focal user-side outcome is a binary indicator for constructive engagement. At the conversation level, we use both the count of constructive user turns and an indicator for whether at least one constructive user turn occurs. Constructive ratios divide constructive user turns by total user turns. Cognitive engagement ratios include passive, active and constructive cognitive engagement.

At the assistant side, scaffolded support is represented by the S2 label. Conversation-level scaffolding presence indicates whether at least one assistant turn in the conversation is labelled S2. Support-form analyses condition on conversations or

turns with scaffolded support and then compare the presence versus absence of each means label. These comparisons are descriptive because means labels can co-occur and because conversations are not randomly assigned to receive a support form.

Behavioural depth is computed from turn counts and turn ordering. Post-answer depth counts turns after the first assistant response. Persistence after failure is analysed only for coding-oriented conversations because debugging failures, error messages and failed-execution contexts are more directly observable in coding than in writing.

A.6 Statistical model specification

The main analyses use simple estimands matched to the claim being made. Descriptive prevalence analyses report proportions within platform-by-task settings. Context analyses compare learning intent, task ecology and depth groups. Conversation-level support associations use Poisson generalized linear models for constructive-turn counts and logistic models for binary constructive-engagement outcomes. Adjacent-turn analyses use assistant-to-user and user-to-assistant turn pairs to estimate local conditional probabilities.

Covariates are included when they correspond to observed compositional differences at the analysis level. Depending on the analysis, these include learning intent, topic or task indicators, turn counts, emotional ratios, error-related variables and turn index. Adjusted estimates are interpreted as measured-covariate-adjusted associations. They do not remove unobserved user motivation, prior expertise, task difficulty or time-varying conversational state.

A.7 Temporal scale

The temporal analyses deliberately separate adjacent-turn ordering from coarser within-conversation summaries. The adjacent-turn analyses ask whether the immediately following user turn is more likely to be constructive after scaffolded support than after a non-scaffolded reference response, and whether that contrast depends on the prior user state. The reverse adjacent-turn analysis asks whether the assistant is more likely to provide scaffolded support after constructive, active or passive user turns.

The coarser analyses ask a different question: whether constructive engagement is higher after the first scaffolded assistant turn than before it, and whether constructive engagement increases from the first to the second half of a conversation. These are not treatment-effect estimates. The first scaffolded turn may occur precisely because the user is struggling, because a difficult task has emerged or because the exchange is already developing. The main temporal claim is therefore scale-specific: the clearest evidence concerns local, state-conditioned coupling rather than cumulative uplift.

Appendix B Robustness and interpretive boundaries

B.1 WildChat model-snapshot robustness

WildChat includes user-facing model metadata that make it possible to repeat major descriptive patterns across model snapshots and coarse model families. These

analyses preserved the main directional patterns reported in the main text: coding generally showed higher cognitive engagement than writing; intentionality gaps were visible across most user-facing models; scaffolded conversations were generally more constructive; and adjacent-turn contrasts were more interpretable than coarser within-conversation contrasts. Supplementary Figures D3–D7 report these WildChat model-snapshot checks.

These checks are used as robustness analyses rather than model-causal comparisons. WildChat model labels are not experimentally assigned. They may be confounded with calendar time, user population, task mix, interface changes and logging context. For this reason, the manuscript does not claim that a specific model family caused the observed engagement differences.

B.2 Label uncertainty

The label scheme intentionally separates top-level constructs from finer support forms. Top-level labels such as constructive engagement and scaffolded support have clearer behavioural anchors. Fine-grained labels, especially sparse or boundary-heavy labels, are less stable. This is why the main claims rely most heavily on constructive engagement, scaffolded support presence, support-form patterns that are consistent across settings and temporal contrasts that can be expressed with simple conditional probabilities.

B.3 Selection into support

The most important threat to causal interpretation is selection into support. More complex tasks, more motivated users or already-developing exchanges may both elicit scaffolded support and produce constructive follow-up. Conversation-level adjusted models reduce measured compositional differences but cannot remove unobserved selection. Adjacent-turn analyses reduce the temporal window and condition on local state, but they still cannot identify randomized exposure effects. The safest interpretation is therefore local, state-dependent coupling between assistant support and user engagement.

B.4 Privacy-preserving reporting

The study analyses naturalistic conversations, so the manuscript avoids reproducing raw conversation text. Operational examples are de-identified and paraphrased to illustrate decision rules without exposing user content. Aggregate statistics, model summaries and validation metrics are reported in the manuscript and supplementary tables.

Appendix C Supporting tables

Table C1 Operational examples for turn-level labels. Examples are de-identified and paraphrased from validation review patterns; they illustrate the decision rule rather than reproducing raw conversation text.

Construct	Paraphrased turn	Annotation rationale
P passive user engagement	"Thanks, that makes sense."	Minimal uptake or acknowledgement without new reasoning, evaluation or task transformation.
A active user engagement	"Can you convert that answer into a command I can run in my setup?"	Task-relevant follow-up or manipulation of supplied information, but without elaborating or testing an idea.
C constructive user engagement	"If the cache expires before the asynchronous callback returns, would the same race condition still happen?"	The user tests an implication, boundary condition or transfer of the prior explanation.
S1 direct support	"Here is the requested function/email draft."	The assistant primarily completes the requested deliverable without regulating the user's process.
S2 scaffolded support	"First isolate the failing step; that will show whether the issue is parsing or state management."	The assistant structures the user's process and leaves diagnostic or revision work with the user.
M1 feedback	"Your diagnosis is close; the bug is in the index offset, not the loop boundary."	Evaluative feedback on the user's prior attempt or understanding.
M2 hinting	"Check the parser log around the first malformed token before changing the model."	A cue or partial path that helps the user proceed without fully resolving the task.
M3 instructing	"First split the condition into two branches, then add a regression test."	Explicit steps or strategy that organizes the user's work.
M4 explaining	"The exception occurs because the event loop is already running, so a nested call cannot acquire it."	A rationale or mechanism that clarifies why the issue occurs.
M5 modeling	"A small pattern you can adapt is: validate input, normalize it, then pass it to the writer."	A sample approach intended to be emulated or adapted rather than merely copied.
M6 questioning	"Which constraint matters more here: preserving tone or shortening the paragraph?"	A question that advances problem solving or reflection.

Table C2 Human–human agreement for labels used in the main analyses. Coding uses the latest three-round validation rerun over 658 task turns, 643 after deduplication; writing uses the latest writing-210 best-pipeline validation. F1, Matthews correlation coefficient (MCC) and Gwet’s AC1 are reported per binary label. Coding MCC and AC1 are recomputed from the exported support and positive-count fields; writing metrics are from the exported agreement report. Rare degenerate labels with no positive cases in a validation set are omitted.

Domain	Role	Label	F1	MCC	Gwet AC1	Support
Coding	User	Active (A)	0.75	0.72	0.87	315
Coding	User	Constructive (C)	0.85	0.82	0.90	315
Coding	User	Passive (P)	0.88	0.87	0.96	315
Coding	User	Emotional (E)	0.83	0.82	0.97	315
Coding	Assistant	Direct support (S1)	0.78	0.73	0.88	328
Coding	Assistant	Scaffolded support (S2)	0.87	0.69	0.70	328
Coding	Assistant	Feedback (M1)	0.90	0.89	0.95	328
Coding	Assistant	Hinting (M2)	0.79	0.75	0.89	328
Coding	Assistant	Instructing (M3)	0.81	0.77	0.92	328
Coding	Assistant	Explaining (M4)	0.86	0.79	0.84	328
Coding	Assistant	Modeling (M5)	0.84	0.79	0.87	328
Coding	Assistant	Questioning (M6)	0.82	0.81	0.95	328
Writing	User	Active (A)	0.38	0.38	0.69	90
Writing	User	Constructive (C)	0.89	0.83	0.86	90
Writing	User	Passive (P)	0.80	0.81	0.99	90
Writing	User	Emotional (E)	0.67	0.65	0.96	90
Writing	Assistant	Direct support (S1)	0.46	0.43	0.86	120
Writing	Assistant	Scaffolded support (S2)	0.97	0.80	0.93	120
Writing	Assistant	Feedback (M1)	0.89	0.80	0.80	120
Writing	Assistant	Hinting (M2)	0.82	0.70	0.71	120
Writing	Assistant	Instructing (M3)	0.71	0.67	0.86	120
Writing	Assistant	Explaining (M4)	0.84	0.70	0.70	120
Writing	Assistant	Modeling (M5)	0.94	0.91	0.95	120
Writing	Assistant	Questioning (M6)	0.96	0.96	0.99	120

Table C3 Analytic model families used in the main analyses. The table summarizes descriptive and conversation-level model families. Adjusted models are interpreted as covariate-adjusted associations in naturalistic data, not as causal treatment-effect estimates.

Analysis family	Unit and outcome	Main contrast or model	Scope
Engagement prevalence	User turns and conversations; cognitive, constructive and emotional engagement indicators.	Descriptive proportions by platform-by-task setting, task ecology and learning intent.	Establishes where learning-oriented engagement is visible; no causal contrast is implied.
Engagement composition	Cognitively engaged user turns; passive, active and constructive depth levels.	Composition of cognitive engagement within each platform-by-task setting.	Motivates constructive engagement as the focal strict outcome while preserving the broader cognitive-engagement hierarchy.
Interaction depth	Conversations; at least one constructive user turn.	Probability of constructive engagement across behavioural-depth buckets.	Depth is interpreted as interactional opportunity and task development, not as a manipulated exposure.
Scaffolding presence	Conversations; constructive-turn count and any constructive turn.	Poisson generalized linear models for counts and logistic models for binary outcomes, comparing conversations with versus without scaffolded assistant turns.	Adjusted estimates reduce measured compositional differences but remain associational because support is selected into by task difficulty and user state.

Table C4 Additional analytic model families and robustness checks. These analyses qualify the main associations by support form, temporal scale and available WildChat model metadata.

Analysis family	Unit and outcome	Main contrast or model	Scope
Support-form associations	Scaffolded conversations; cognitive or constructive engagement ratios.	Mean differences for conversations containing a given support-means label versus scaffolded conversations without that label, stratified by learning intent.	Descriptive because means labels are non-exclusive and can co-occur in the same assistant turn.
Local temporal coupling	Adjacent assistant–user or user–assistant turn pairs; next constructive user turn or next scaffolded assistant turn.	Conditional probability contrasts around adjacent turns; detailed estimands are listed in Supplementary Table C5.	Interpreted as local ordering and state-dependent coupling, not as randomized support effects.
Coarse temporal contrasts	Conversations containing scaffolded support; constructive engagement before and after the first scaffolded turn.	Mean after-minus-before contrast and first-half versus second-half robustness summaries where available.	Tests longer-span accumulation patterns, but the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
WildChat model robustness	WildChat conversations or turns; main prevalence, association and temporal outcomes.	Repetition across user-facing model snapshots or coarse model families.	Robustness summaries only; model labels are observational and may be confounded with time, user mix and task mix.

Table C5 Temporal estimands used to separate local coupling from coarse accumulation. The analyses intentionally combine adjacent-turn and longer-span summaries because the theoretical claim concerns temporal scale. Adjacent-turn contrasts test local ordering; within-conversation contrasts test whether there is evidence for broader accumulation after scaffolding appears.

Estimand	Definition	Reported in main text	Scope
Overall adjacent-turn lift	Difference in the probability that the next user turn is constructive after scaffolded support versus after a non-scaffolded reference response, estimated within each platform-by-task setting.	Figure 5a	Local association between assistant support type and immediately following constructive engagement; not a randomized support effect.
Prior-state adjacent contrast	Difference in next-turn constructive probability after scaffolded support versus reference response, separately for prior constructive, active and passive user states.	Figure 5b	Tests whether local support–engagement coupling depends on the user’s immediately preceding state.
Reverse state-conditioned support	Probability that the next assistant turn is scaffolded, estimated after constructive, active and passive user turns.	Figure 5c	Describes how assistant scaffolding supply varies with the user’s current engagement state.
Around-first-scaffold contrast	Mean constructive engagement after the first scaffolded assistant turn minus mean constructive engagement before that turn, among conversations containing scaffolded support.	Figure 5d	Coarse before–after contrast; the first scaffolded turn may be selected by difficulty, uncertainty or an already developing exchange.
First-half versus second-half contrast	Constructive engagement in the second half of a conversation minus constructive engagement in the first half, reported where available in WildChat.	Results text	A longer-span robustness lens; increases do not identify scaffolding as the cause of later engagement.

Table C6 Support-supply profile within scaffolded assistant turns. Values show the percentage of scaffolded assistant turns containing each scaffolding-means label. Means labels are not mutually exclusive, so rows do not sum to 100%. Coding-oriented scaffolded turns are dominated by explanation-heavy support, whereas writing-oriented scaffolded turns are dominated by instructing support.

Setting	M1 Feedback	M2 Hinting	M3 Instructing	M4 Explaining	M5 Modelling	M6 Questioning
WildChat coding	8.8	14.7	29.7	84.0	23.9	6.9
WildChat writing	16.6	9.6	59.2	29.5	15.0	6.9
Copilot coding	10.1	21.5	21.4	77.5	24.5	6.6
Copilot writing	9.4	9.8	66.5	28.8	11.4	4.2

Appendix D Supporting figures

The supporting figures report extended temporal checks and WildChat model-snapshot robustness analyses. They are included to make the main claims auditable without shifting the manuscript’s emphasis away from the four matched dataset-by-task settings. Model-snapshot figures are descriptive robustness checks because WildChat model labels are observational rather than experimentally assigned.

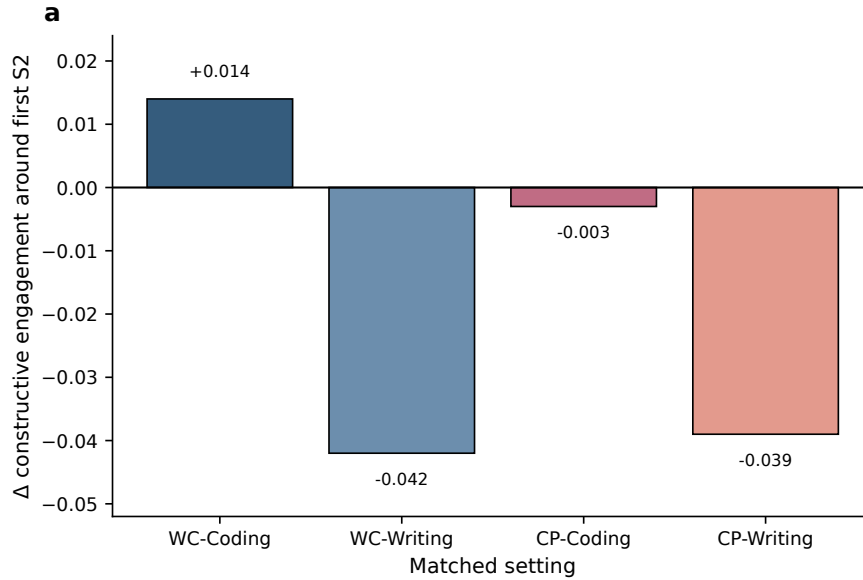


Fig. D1 Coarse around-first-scaffold contrasts provide limited evidence for cumulative improvement. Bars show the difference in constructive engagement after versus before the first scaffolded assistant turn within conversations containing scaffolded support. The contrasts are small and mixed across dataset-by-task settings, supporting the main-text interpretation that the clearest temporal evidence concerns local adjacent-turn coupling rather than broad cumulative uplift.

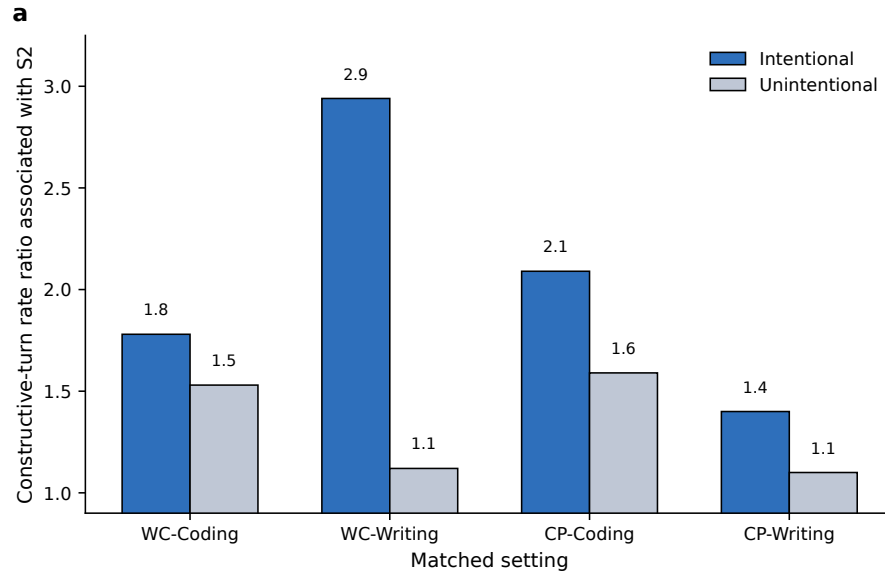


Fig. D2 Scaffolded-support associations vary by user framing. Stratified model estimates compare the association between scaffolded-support presence and constructive participation in intentional versus unintentional conversations. The figure is interpreted as descriptive moderation rather than evidence that intent causally changes support effectiveness, because user framing, task complexity and persistence are not randomly assigned.

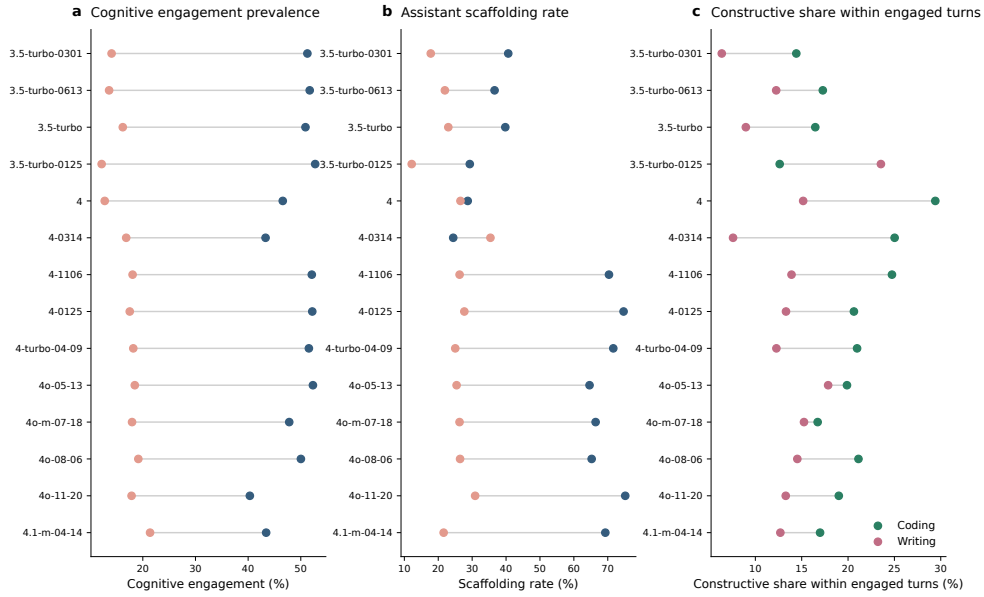


Fig. D3 WildChat model-snapshot prevalence profiles preserve the main domain contrast. Across user-facing WildChat model snapshots, coding-oriented conversations generally show higher cognitive engagement and higher assistant scaffolding rates than writing-oriented conversations. The model labels are observational and may reflect time, user mix and traffic composition, so the figure is used as a robustness check rather than a model-causal comparison.

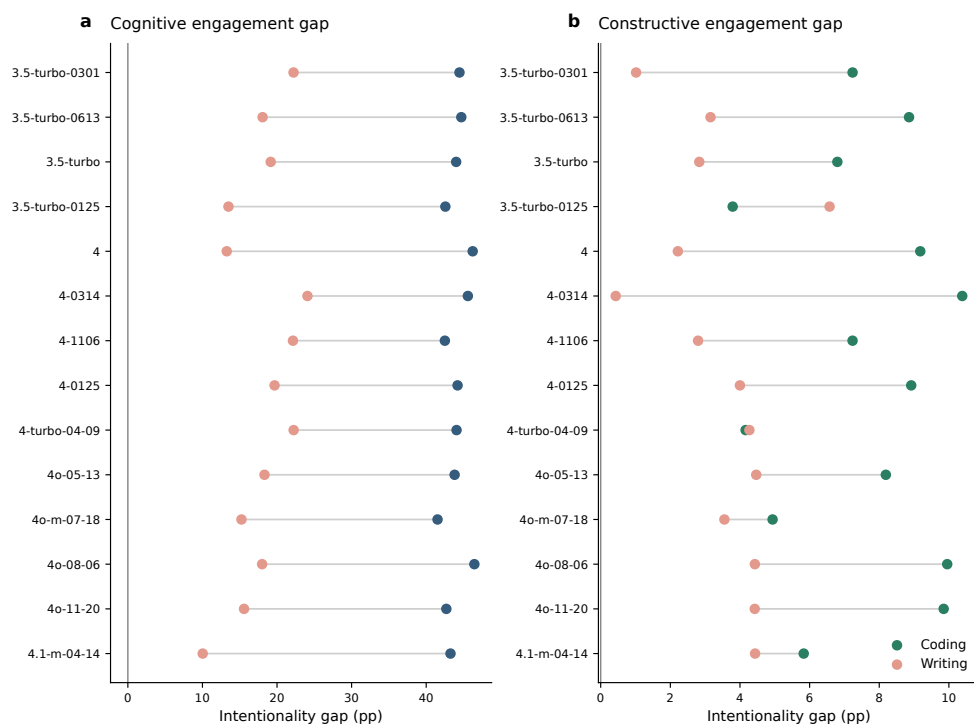


Fig. D4 Intentionality gaps are visible across WildChat model snapshots. The figure reports model-specific differences between intentional and unintentional conversations in cognitive and constructive engagement, separately for coding and writing. Most model snapshots preserve the directional pattern that intentionally framed conversations contain richer learning-oriented participation.

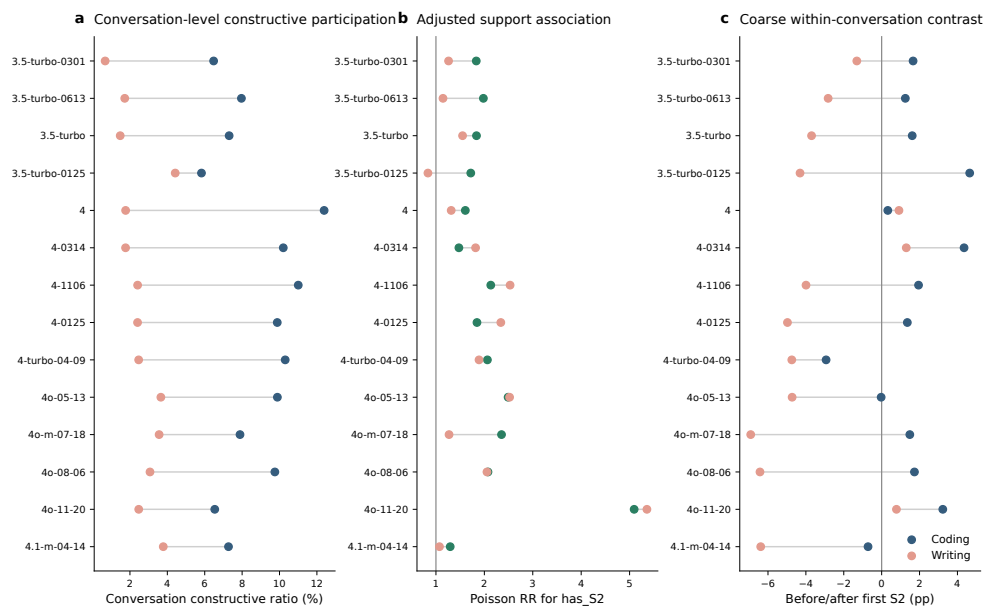


Fig. D5 Conversation-level support associations are broadly positive across WildChat model snapshots. The figure repeats major conversation-level analyses by user-facing WildChat model snapshot, including constructive participation, adjusted support associations and coarse before–after contrasts around the first scaffolded assistant turn. The pattern supports the distinction between stable conversation-level associations and more mixed coarse temporal contrasts.

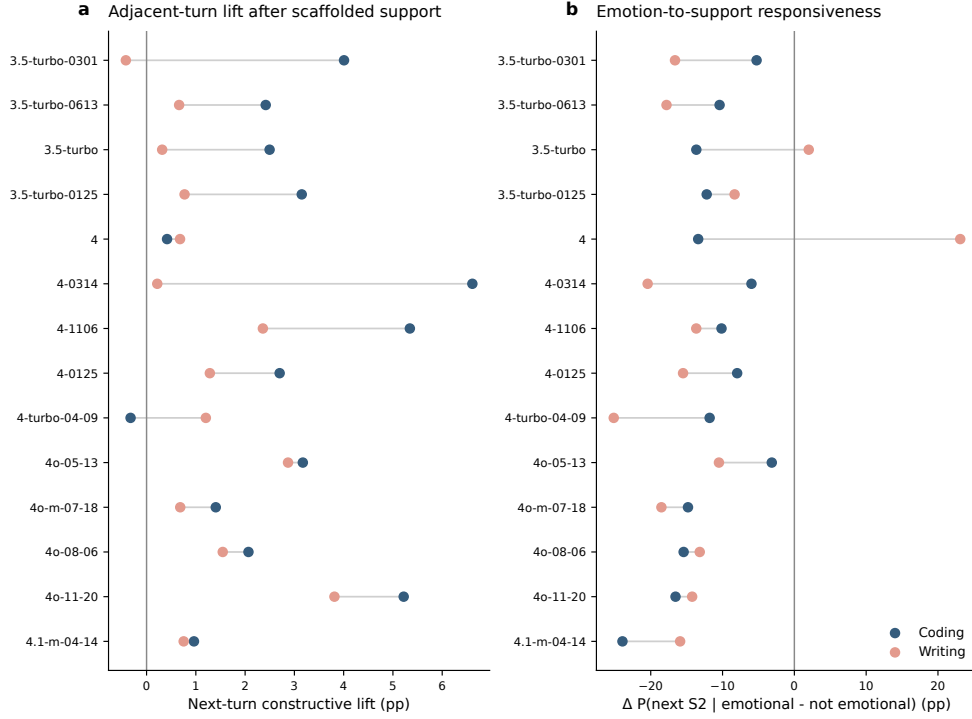


Fig. D6 Local temporal signals vary across WildChat model snapshots. The figure reports adjacent-turn constructive lifts and emotion-to-support responsiveness by model snapshot. Adjacent-turn constructive lifts are often positive, especially in coding-oriented conversations, but the magnitude varies across snapshots; emotion-to-support responsiveness is more heterogeneous and is treated as exploratory.

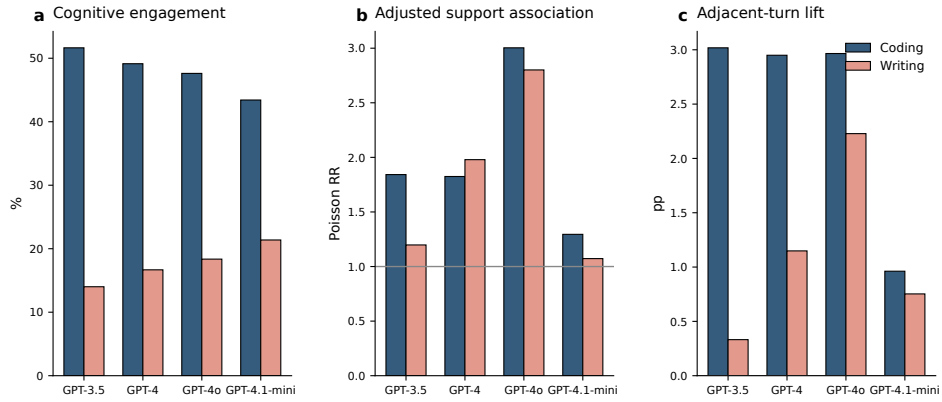


Fig. D7 Coarse model-family aggregation preserves the main directional patterns. WildChat model snapshots are grouped into coarse model families to summarize cognitive engagement prevalence, adjusted support associations and adjacent-turn constructive lift. Coding remains higher than writing across families, and the support-association and local-lift patterns remain directionally consistent on average.